

Statistical and Computational Models

Lecture 0: Course Introduction

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And suppose we fit just a single model:

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where $\mathbf{x}_i$ collects all the other covariates.

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Can we answer all three questions with the same data and model? Maybe...

Three goals

Question	The target	Judged by
Descriptive	a statement about a population	does it recover the quantity?
Causal	a contrast between potential outcomes	plausible assumptions?
Predictive	a projection about an unrealised variable	error on unseen data?

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The **estimand**: the target of inference.

What good is a regression? (Spirling & Stewart, 2025)

The strength of any answer we give to any empirical question depends on:

- The coherence of the question (and estimand)
- The quality of the explanation we develop
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But note: A fancy model does not (in any way, shape, or form) guarantee a coherent question, a good explanation, or a strong connection between your evidence and those things.

Course outline

The course covers four units:

Unit	Lectures	Dates
1 · Regression as statistical learning	1–5	Aug 25 – Sep 8
2 · Likelihood, GLMs, and Bayesian computation	6–11	Sep 10 – Sep 29
3 · Learning beyond the linear model	12–17	Oct 1 – Oct 22
4 · Non-standard data	19–27	Oct 29 – Dec 1

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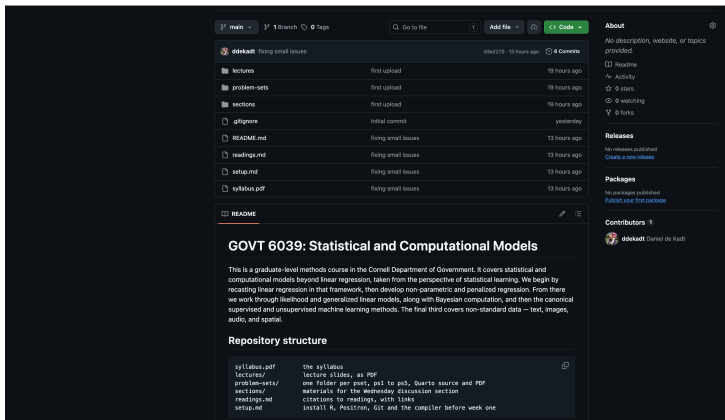
Units 1–3 build a statistical and computational **foundation and core**.

Unit 4 considers **non-standard data** that are becoming increasingly common in computational social science.

Course website

The course will principally be run through GitHub, from <https://github.com/ddekadt/cornell-govt-6039>.

github.com/ddekadt/cornell-govt-6039



The screenshot displays the GitHub repository page for `ddekadt/cornell-govt-6039`. The repository is on the `main` branch. The file list on the left includes:

- `lectures`: first upload, 19 hours ago
- `problem-sets`: first upload, 19 hours ago
- `sections`: first upload, 19 hours ago
- `gfigs`: initial commit, yesterday
- `README.md`: fixing small issues, 13 hours ago
- `readings.md`: fixing small issues, 13 hours ago
- `setup.md`: fixing small issues, 13 hours ago
- `syllabus.pdf`: fixing small issues, 13 hours ago

The `README` file is selected, showing the following content:

GOVT 6039: Statistical and Computational Models

This is a graduate-level methods course in the Cornell Department of Government. It covers statistical and computational models beyond linear regression, taken from the perspective of statistical learning. We begin by recasting linear regression in that framework, then develop non-parametric and penalized regression. From there we work through likelihood and generalized linear models, along with Bayesian computation, and then the canonical supervised and unsupervised machine learning methods. The final third covers non-standard data — text, images, audio, and spatial.

Repository structure

```
syllabus.pdf      the syllabus
lectures/         lecture slides, as PDF
problem-sets/    one folder per pset, p1 to p4, Quarto source and PDF
sections/        materials for the Wednesday discussion section
readings.md      citations to readings, with links
setup.md         install R, Positron, Git and the compiler before week one
```

Lecture and discussion section

LECTURE

Tue and Thu, 75 minutes.

Device-free: no laptops, no phones.

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Device-free: no laptops, no phones.

DISCUSSION SECTION

Wed, 50 minutes, TA-led.

Bring a laptop, leave your phone at home.

Problem sets

Five, released bi-weekly, featuring a mix of theory, simulation, and replication of published work. But they are **ungraded**!

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Do not hand them to a model. You will be wasting your own time and valuable tokens.

Assessment

Component	Weight	When
Problem sets (five, ungraded)	0%	Bi-weekly; see the calendar
Mid-term, in class	40%	In class, Tue Oct 27
Reading presentation	10%	Assigned in week 2
Final project	40%	Paper and code due [DATE TBD]
Project presentation	10%	In class, Thu Dec 3

Mid-term

The mid-term will take place in class on Tue Oct 27. You will have 75 minutes.

It will cover units 1–3. You may bring two pages of notes, hand-written or printed.

No devices of any kind.

Final project

You will produce a paper of 2,000–4,000 words, plus a reproducible code repository, tackling an empirical problem of your choice, using methods covered in this course.

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Co-author with another student if you like, but co-authoring with faculty needs permission in advance.

Project milestones

Milestone	When
Candidate questions and data identified	September
Check-in meeting with the instructor	before Tue Oct 6
Presentations and peer review	Lecture 28 · Thu Dec 3
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The check-in is a meeting about **feasibility** of your project. Please come prepared (i.e., have looked for data already).

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- **ISL** – James, Witten, Hastie & Tibshirani (2021).

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- Four more recur: **CASI** (Efron & Hastie), **SLP** (Jurafsky & Martin) **GCR** (*Geocomputation with R*), **SDS** (*Spatial Data Science*).

Software

The course will be primarily taught in **R**, but we will also use **Python** at points.

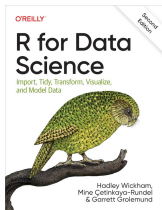
All write-ups should be built as reproducible documents using **Quarto**, and I encourage you to practice version control in **Git**.

I recommend you consult these (free online) books about **R** and **Python**:

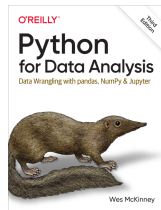
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r4ds.hadley.nz



wesmckinney.com/book

Artificial intelligence

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But it is really bad for your learning. So, here is my vibe:

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Remember: you are responsible for work submitted in your name, even if Claude or Codex has written some of it.